i An update to this article is included at the end

EARTH AND PLANETARY SCIENCE LETTERS 6 (1969) 349-354. NORTH-HOLLAND PUBLISHING COMP., AMSTERDAM

PALAEOMAGNETIC RESULTS FROM SOME BRITISH TERTIARY DYKES

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Received 2 April 1969
Revised manuscript received 24 June 1969

Nine Tertiary dykes have been sampled at eleven sites and found to be reversely magnetised. Several explanations of the shallow inclinations of the direction of magnetisation are considered but at present there is insufficient evidence to decide between them.

1. Introduction

When this collection was made all the Tertiary lavas and all but two of the dykes from Great Britain which had been studied had been shown to be magnetised in the reverse sense [1–12]. Much of the earlier work was done without either alternating field or thermal demagnetisation and so far absolute ages have been found for very few of the rocks studied palaeomagnetically.

The work of Heirtzler et al. [13] implies that there have been many intervals of normal polarity in the last 76 Ma (Mega-année = 10^6 years [14]) so the British rocks apparently represent one or more short periods, rather than a single long epoch, when the geomagnetic field was reversed.

This report describes work which forms part of an extensive reappraisal of the palaeomagnetic results from the British Tertiary province.

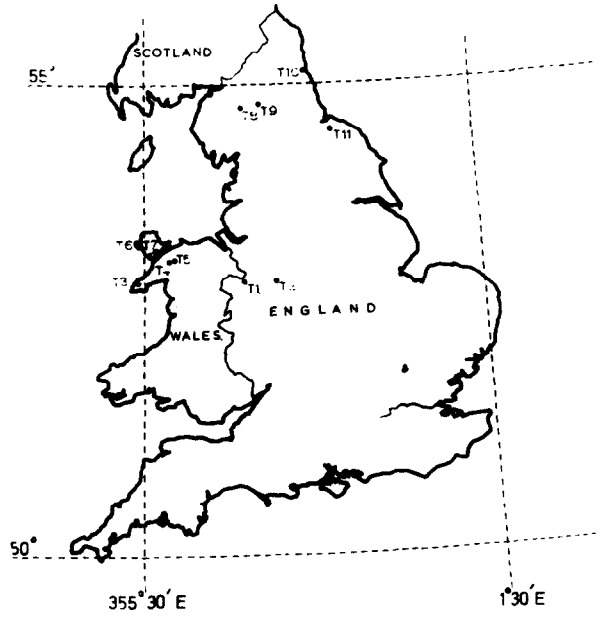


Fig. 1. Outline map of England and Wales showing location of collecting sites.

2. Sampling

Samples of Tertiary dykes have been collected from eleven different sites in various parts of England and Wales. Three sites, T8, 9 and 11, are thought to be different exposures of a single dyke: all the others are distinct formations. Between 4 and 6 independently oriented cores were obtained from each exposure using a portable drill. Because some of the exposures were in hollows so that no distant geographic points could be seen and because of the notorious British weather it was only possible to take satisfactory geographic bearings for three of the dykes, and an approximate one for a fourth. Magnetic bearings were obtained for all the cores. For the three dykes, where sighting was possible, T5, T8 and T9, the magnetic and geographic bearings agreed to less than 4°

for each of the eleven individually oriented cores and the average divergence was 1.2°. In this analysis the magnetic bearings have been used.

A single core was also collected from the country rock baked by dyke T2. This was the only case where it was possible to sample the baked rock. Fig. 1 shows the location of the sites; map references and geological notes are given in the appendix.

3. Measurements

All measurements were made with an astatic magnetometer. After initial measurement a specimen from one core from each site was demagnetised successively in eight AC fields 5, 10, 15, 20, 25, 30, 35 and 40 milli-Tesla (10^{-3} Wbm^{-2} or 10 Gauss) [14] and remeasured after each step.

Most of the samples proved to be relatively unstable during alternating field demagnetisation, either direction or intensity changing rapidly with increasing demagnetising field. This behaviour appears to be typical of dykes. The results for these 11 "special" specimens suggested that the optimum field to give the greatest removal of viscous components together with least scattering lay between 15 and 20 mT. A specimen from each of the other cores was measured after demagnetisation at 0, 15 and 25 mT. The mean directions for the samples from each dyke were found for both the 15 and 25 mT sets of results; the set with the smallest scatter being chosen in each case. In only one case was this the 25 mT set.

Errors in measurement were small (< 2°) and most of the uncertainty arises from the unavoidable use of magnetic orientation. There may be a further uncertainty due to tectonic movement as it was not possible to estimate the amount of tilting, if any. The final errors of the unit mean directions are given by the angular standard deviation, δ, and the error in the mean, ϵ, shown in the table.

The agreement between the baked rock collected at Butterton and the dyke is quite good and confirms the shallow inclination.

The mean directions for T5 and T7 are very nearly the same, this may indicate a similar age for the two dykes.

An attempt was made at Blunderfield, T8, to verify the variations across the dyke reported by Bruckshaw and Robertson [1]. The five samples we collected were taken from different parts of the dyke yet they all gave similar directions of magnetisation after cleaning, and the α_{95} for the mean direction is less

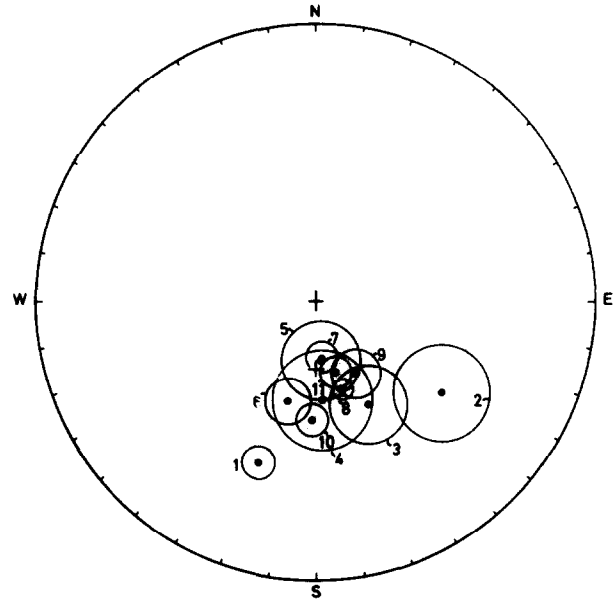


Fig. 2. Equal area projection showing mean directions for the 11 sites. The 95% confidence limits are shown by the circles; + is the direction of the field due to a centred axial dipole.

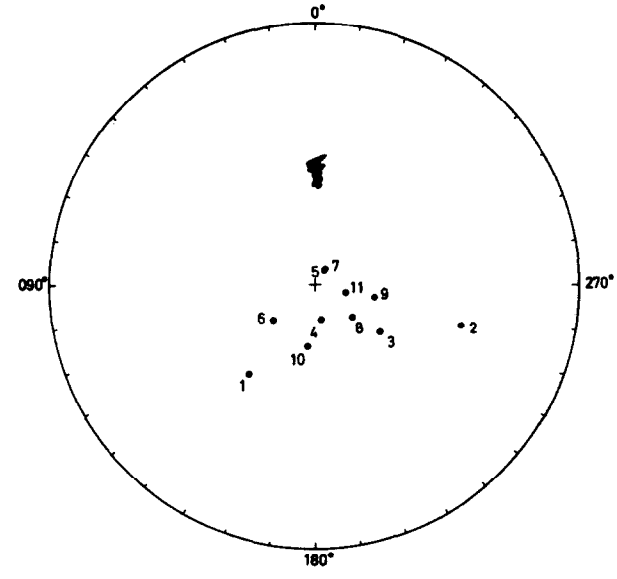


Fig. 3. Equal area projection with North Geographic Pole at centre showing virtual Pole positions.

Table 1
Mean directions of magnetisation, virtual pole positions and associated statistics.

Code	Location	Lat. °N	Long. °E	N	R	D °ETN	I °down	δ deg	ϵ deg	k	α_{95} deg	γ deg	Polarity	P_{lat} °N	P_{long} °E	δp deg	δm deg
T 1	Clive nr. Wem, Shropshire	52.81	357.3	4	3.9875	199.57	−38.46	4.54	2.27	239.1	5.95	32	R	55	144	4	7
T 2	Butterton nr. Stone, Staffordshire	52.98	357.75	5	4.8482	126.62	−43.42	14.15	6.33	26.36	15.18	37	R	42	254	12	19
B 2	Idem			1	1.0000	123.24	−39.18						R	37	255		
T 3	Penrhyn Colman Caernarvonshire	52.88	355.32	4	3.9530	153.74	−55.50	8.79	4.40	63.83	11.59	18	R	65	233	12	17
T 4	Marchlyn Mawr	53.13	355.93	4	3.9198	176.19	−60.89	11.50	5.75	37.39	15.22	9	R	79	190	18	23
T 5	Ogwen Cottage	53.12	355.97	4	3.9531	175.44	−72.25	8.78	4.39	63.95	11.58	3	R	85	326	18	20
T 6	Porth Dafarch Anglesey	53.28	355.35	6	5.9496	195.47	−59.25	7.43	3.03	99.21	6.76	12	R	73	131	8	10
T 7	Gwydryn Quarry Anglesey	53.19	355.75	5	4.9877	174.45	−72.84	4.02	1.80	325.87	4.24	4	R	84.01	326.61	7	8
T 8	Blunderfield quarry Nr. Penrith	54.78	357.33	5	4.9948	163.53	−62.48	2.62	1.17	767.13	2.76	10	R	74.73	228.30	3	4
	(McMurry)			22	21.238	158.6	−64.3	–	–	27.57	6.0		R				
T 9	Tynehead Snae Alston Moor	54.72	357.63	6	5.9405	151.78	−65.53	8.08	3.30	83.96	7.35	12	R	71.13	257.47	10	12
T 10	Collywell Bay Seaton Sluice	55.08	358.54	6	5.9776	182.02	−54.88	4.95	2.02	223.70	4.49	16	R	70.28	173.66	5	6
T 11	Gt. Ayton Yorkshire	54.50	358.88	5	4.9871	165.00	−68.07	4.12	1.84	309.85	4.35	6	R	80.36	254.75	6	+7
Cleveland – Armathwaite Dyke T8, 9, 11				3	2.9943	160.08	−65.47	3.35	1.93	391.1	6.24		R	–	–	–	–
Overall mean, Dykes T1–11				11	10.6132	169.53	−61.38	15.24	4.59	25.9	9.15	10	R	–	–	–	–

N = number of cores collected and measured
R = length of the resultant of the N unit vectors
D = mean declination of remanent magnetisation (ETN)
I = mean inclination of remanent magnetisation (+ve down)
$\delta = \cos^{-1}(R/N)$ angular standard deviation [7]
$\epsilon = \delta/\sqrt{N}$ angular standard deviation of mean [7]
k = estimate of precision parameter [20]
α_{95} = semi-angle of core of 95% confidence for mean direction [20]
γ = angle between mean direction and dipole field
δp, δm = semj-axes of oval of confidence for virtual pole position

than 3°. An earlier collection of oriented lumps from the same site has been measured by E.W.McMurry at Imperial College, London (private communication). The results of his thermal demagnetisation experiments on 22 independently oriented specimens spread across the whole dyke are included in table 1. These results suggest that Bruckshaw and Robertson's earlier result was due to secondary magnetisation.

T8, T9 and T11 are believed to be parts of a single dyke running for about 120 km across the whole breadth of Northern England. If this is so then the directions of magnetisation found for the three sites sampled should be in close agreement. This does indeed appear to be the case; the between site mean direction is −65° at 160° ETN with $\alpha_{95} = 6°$ so that even if the exposures are not part of a single dyke it seems probable that they were all intruded at about the same time.

All the sites lie between 52.8°N and 56°N and a centred axial dipole field would give directions between −69° and −71° at 180° ETN. It is noteworthy that many of the inclinations are much less; none are significantly greater. In all, 8 out of the 11 do not include the dipole field direction within the circle of confidence and each of these has a shallower dip than the dipole field.

4. Discussion

There could be several different explanations for the difference between the directions of magnetisation and the dipole field.

4.1. *Continental drift*

If the site of a dyke intrusion had moved northward since the time of magnetisation the inclination will be shallower than that expected for the present latitude.

The ancient latitude based on the mean direction is 12° different from the mean of the present latitudes. Continental movement at an average rate of 20 mm/a would produce a change of this order in about 70 Ma. While such a movement cannot be ruled out it seems unlikely. However, the circle of confidence about the mean direction is itself just sufficient to explain the divergence so that some continental movement together with other effects could explain the results. Precise dating of the rocks would test the feasibility of such movements.

4.2. *Tectonic tilting*

It is possible that during the active periods of the Tertiary the dykes have been moved from their original attitudes.

The amount of tilt required to bring the mean direction of the 11 dykes into coincidence with the dipole field is 10° at 330° ETN. Such a large systematic tilting seems unlikely with such scattered locations.

4.3. *Transitional field*

Several instances have been reported where a sequence of specimens has directions of magnetisation changing systematically from one polarity to the other [15]. It is possible that by chance we have sampled dykes which were magnetised during one or more such transitions.

The estimated time for a complete transition is of the order of 10 ka (10 kiloannee) [16–18]. The rocks sampled could have been magnetised any time during a period of the order of 50 Ma. According to Heirtzler et al. [13] there were 90 transitions between 25 Ma and 75 Ma ago so that the field could have been in an intermediate state 1.8% of the time. It is improbable that all the dykes were magnetised at these times but precise dating is required before any final conclusions can be made.

4.4. *Semi-permanent non-dipole field*

If in addition to the main dipole field there was a local non-dipole field which persisted, at least statistically, for a long time, the remanent magnetisation would be affected. If we assume the field to be in a meridional plane then a non-dipole component with an inclination in the south-down arc between the normal and reversed directions of the dipole field would cause reversely magnetised rocks to have shallower and normally magnetised rocks to have steeper inclinations than expected. The effect of such a non-dipole field is distinguishable from the effect of tilting and continental movement by the different effect it has on normal and reversed samples of the same age.

Shallow reversed directions are reported in all the published investigations [1–12], although only

Raja [11] and Smith [12] carried out any systematic cleaning of their specimens. The only reliable normal directions for British Tertiary rocks are also reported by Raja [11] and Smith [12]. The direction for the dyke and the mean direction of six baked specimens reported by Raja are steeper than the dipole field. Of the three specimens from the dyke reported by Smith two have shallow inclinations and one a steeper inclination than expected. The two specimens from the associated baked sandstone have shallower inclinations.

To test this hypothesis it is essential to have a very large collection including both normal and reversed samples. The published work does not form a large enough set and the present collection is far from adequate and contains no normally magnetised rocks. A large collection of dykes has since been made by Mull [19]. Of 165 dykes, 40 are normally magnetised with a mean direction (+77° at 332° ETN) which is steeper than the dipole field and 125 are reversed with a mean direction (−62° at 179° ETN) which is shallower than expected.

5. Conclusion

For the purpose of determining the polarity any errors arising from effects discussed in the previous section or inaccuracies in the magnetic bearings are certainly negligible. The results confirm that these dykes are reversely magnetised.

While it is unlikely that either continental drift, tectonic tilting or a transitional field by themselves can explain the shallow inclinations it is conveivable that a combination of all three with or without non-dipole fields might. However, when the previously published results and the present ones are considered together it is clear that the possibility of non-dipole effects must be examined more closely. For this purpose a very large collection of samples will be necessary and this has already been started in the British Tertiary Province. This extensive collection may reveal the cause of the shallow reversed inclinations although it will be essential to have precise and accurate dating to decide between the possible mechanisms.

Appendix

Map references given with the following geological notes give the sheet numbers and coordinates for the 2½ series of maps published by the Ordnance Survey of Great Britain.

T1 Clive dyke, near Wem, Shropshire (SJ52; 527238); a decomposed porphyritic dolerite intruding both Bunter and Keuper divisions of the Trias. (R.W.Pocock and D.A.Wray, "The geology of the country around Wem", Memoirs of the Geological Survey of England and Wales (1925) p. 62.)

T2 Butterton dyke, near Stone, Staffordshire (SJ84; 834420); an olivine dolerite dyke cutting and baking upper Carboniferous Keele sandstones. (W.Gibson and C.B.Wedd, "The geology of the Country around Stoke on Trent", Memoirs of the Geological Survey of England and Wales (1905) p. 55.)

T3 Penrhyn Colmon, Caernarvonshire (SH23; 193342); one of three NW-SE Tertiary dykes in this area. (C.A. Matley, "The pre-Cambrian complex and associated rocks of south western Lleyn", Quart. J. Geol. Soc. 84 (1928) 440.)

T4 Marchlyn Mawr dykes, Caernarvonshire (SH66; 614618).

T5 Ogwen Cottage dyke, Caernarvonshire (SH66; 644605). These are olivine dolerites which cut lower Palaeozoic (Llandilo) Strata. (D.Williams, "The geology of the country between Nant Peris and Nant Ffrancon", Quart. J. Geol. Soc. 86 (1930) 191.)

T6 Porth Dafarch dyke, Holy Island, Anglesey (SH27; 233799); an ophitic olivine dolerite cutting pre-Cambrian of the Mona Complex. (E.Greerly, "The geology of Anglesey", vol. 2, Memoirs of the Geological Survey of England and Wales (1919) p. 692.)

T7 Gwydryn dyke Anglesey (SH46; 497682); a rather fine basalt with brownish glass. (E.Greerly, op. cit. p. 694.)

T8 Blunderfield quarry, near Kirkoswald, Cumberland (NY54; 564438); cuts Carboniferous and Triassic rocks.

T9 Tynehead Snae, Alston Moor (NY73, 758366). (T.Eastwood, "Northern England", British Regional Geology series of D.S.I.R. (1953) 55, 62.) See T11.

T10 Blyth Swarm, Seaton Sluice, Collywell Bay, Northumberland (NZ37; 340767). Cuts Carboniferous rocks. (J.J.Teall, "Petrological notes on some North of England dykes", Quart. J. Geol. Soc. 40 (1884) 209.)

T11 Cliff Ridge Wood, near Great Ayton, Yorkshire (NZ51; 571118). Cuts Jurassic rocks. (G.Barrow, "The geology of North Cleveland", Memoirs of the Geological Survey of England and Wales (1888) p. 60.)
T8, 9 and 11 are all part of the Cleveland-Armathwaite dyke and are augite, andesite and tholeiite.

Acknowledgements

Thanks are due to Dr. J.A.Miller of the Department of Geodesy and Geophysics at Cambridge, and Dr. F.J.Fitch of Birkbeck College, London, for their help in collecting the samples and to my colleagues in Liverpool for their advice and encouragement.

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Update

Earth and Planetary Science Letters
Volume 8, Issue 6, July 1970, Page 454

DOI: https://doi.org/10.1016/0012-821X(70)90154-8

EARTH AND PLANETARY SCIENCE LETTERS 8 (1970) 454. NORTH-HOLLAND PUBLISHING COMPANY

ERRATA

Effect of strain on fission-track ages of tektites, by S.A. Durrani and D.A. Hancock, Earth and Planetary Science Letters 8 (1970) 157–162.

(i) On page 161, in table 2, the heading of column 2 should read $10^{15}cm^{-2}$ as units for neutron fluence (and not $10^{-15}cm^{-2}$).

(ii) In the same table, the first entry in column 3 should read 7.40 ± 0.35 (and not 7.40 ± 0.035).

Palaeomagnetic results for some British tertiary dyk by Peter Dagley, Earth and Planetary Science Letters 6 (1969) 349-354.

The latitude and longitude of the virtual poles given in table 1 and figure 3 correspond to the intersection of the dipole axis with the northern hemisphere. The Virtual Geomagnetic (North) Pole is of course in the Southern Hemisphere for these reversely magnetised dykes and is found by changing the sign of the latitude and subtracting 180° from the longitude.